

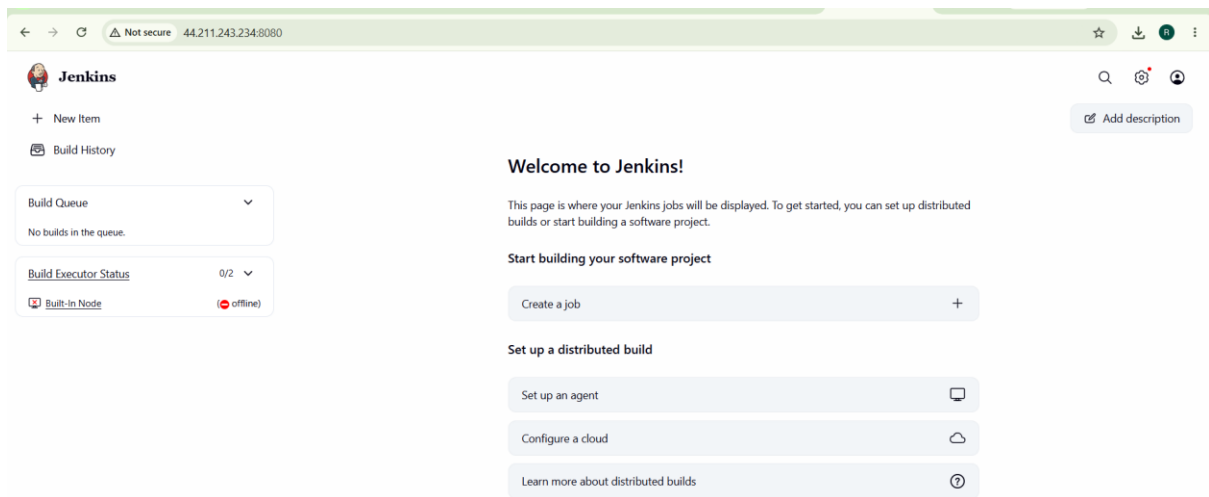
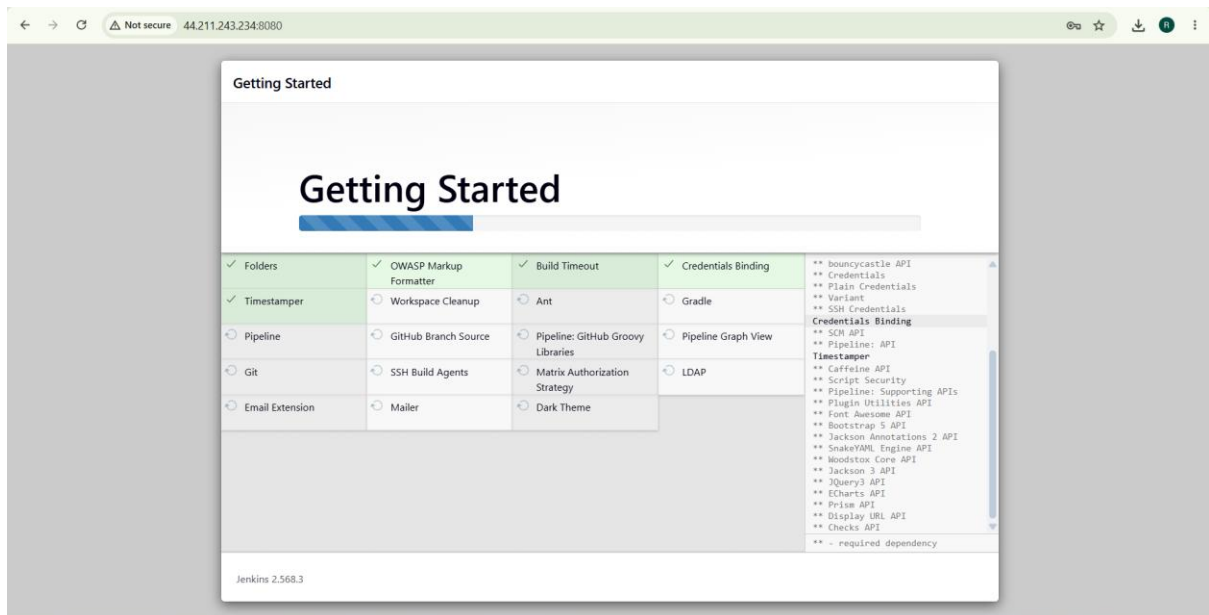
1. Install Jenkins and run Jenkins on port number 8081.

Implementation Steps

- Launch an Amazon Linux 2023 EC2 instance in AWS.
- Connect to the EC2 instance using SSH.
- Install Java 21: `sudo dnf install java-21-amazon-corretto -y`
- Add the Jenkins repository:
`sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo`
- Import the Jenkins repository key:
`sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key`
- Install Jenkins:
`sudo dnf install jenkins -y`
- Configure Jenkins to use port 8081:
`sudo systemctl edit jenkins`
Add:
[Service]
Environment="JENKINS_PORT=8081"
- Reload the systemd configuration:
`sudo systemctl daemon-reload`
- Start Jenkins:
`sudo systemctl start jenkins`
- Enable Jenkins to start automatically after reboot:
`sudo systemctl enable jenkins`
- Check Jenkins service status:
`sudo systemctl status jenkins`
- Allow TCP port 8081 in the EC2 Security Group.
- Access Jenkins using:
`http://<EC2-Public-IP>:8081`
- Complete the initial Jenkins setup and create the Jenkins administrator account.

Result

- Jenkins was successfully installed and configured to run on port 8081.



```
[root@ip-172-31-6-74 ~]# sudo systemctl edit jenkins
[root@ip-172-31-6-74 ~]# sudo systemctl daemon-reload
[root@ip-172-31-6-74 ~]# sudo systemctl restart jenkins
[root@ip-172-31-6-74 ~]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/jenkins.service.d
            -override.conf
   Active: active (running) since Tue 2026-09-29 10:43:52 UTC; 3s ago
     Main PID: 28176 (java)
       Tasks: 39 (limit: 1059)
      Memory: 264.2M
         CPU: 19.066s
    CGroup: /system.slice/jenkins.service
            └─28176 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8081

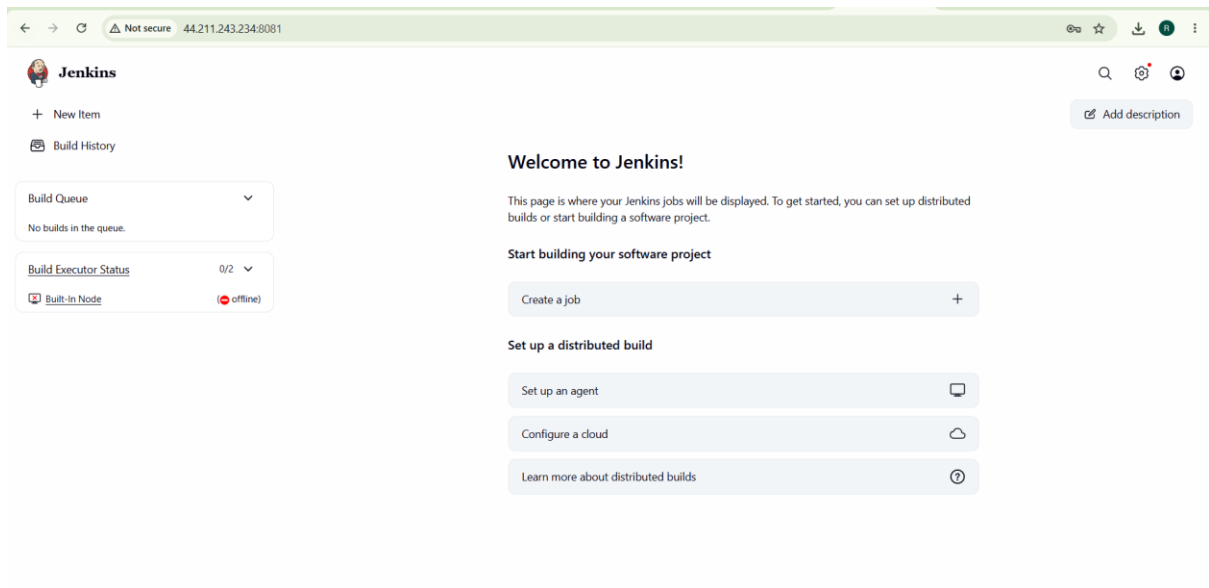
Sep 29 10:43:50 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:50.123-0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
Sep 29 10:43:50 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:50.124-0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
Sep 29 10:43:50 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:50.959-0000 [id=32] INFO h.p.b.g.GlobalTimeoutConfigurationLoad: global timeout not set
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.175-0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.176-0000 [id=33] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.178-0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.183-0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.202-0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal jenkins[28176]: 2026-09-29 10:43:52.340-0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Sep 29 10:43:52 ip-172-31-6-74.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
[root@ip-172-31-6-74 ~]# sudo ss -ltnp | grep 8081
LISTEN 0      50          *:8081     :*      users:((("java".pid=28176,fd=9))
[root@ip-172-31-6-74 ~]#
```

```
## Editing /etc/systemd/system/jenkins.service.d/override.conf
## Anything between here and the comment below will become the new contents of the file

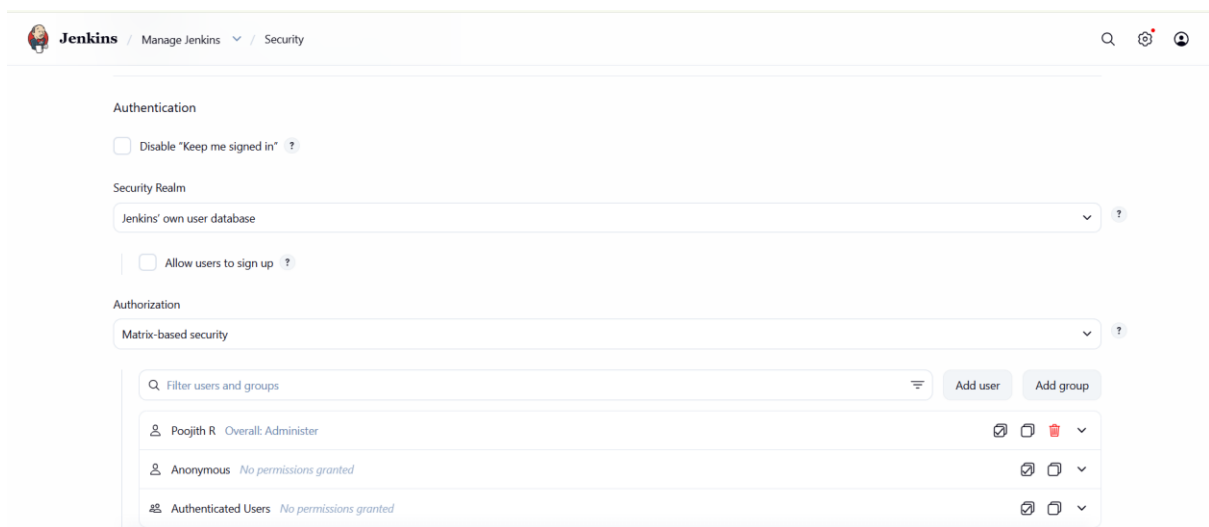
[Service]
Environment="JENKINS_PORT=8081"

## Lines below this comment will be discarded

## /usr/lib/systemd/system/jenkins.service
##
## This file is managed by systemd(1). Do NOT edit this file manually!
## To override these settings, run:
##
##   systemctl edit jenkins
##
## For more information about drop-in files, see:
```



2. Secure Jenkins server



3. Create users called DevOps, Testing in Jenkins with Limited access.

Implementation Steps

- Open: Jenkins → Manage Jenkins → Users
- Click: Create User
- Create the first user:
Username: DevOps
- Set the required password and user details.
- Create the second user:
Username: Testing

- Set the required password and user details.
- Go to: Manage Jenkins → Plugins
- Install Role-Based Authorization Strategy
- Go to: Manage Jenkins → Security
- Open the Role-Based Authorization Strategy configuration.
- Configure roles and assign the required permissions to the users.
- Provide only the required Jenkins permissions rather than full administrative access.

The screenshot shows the Jenkins 'Manage Jenkins' > 'Plugins' page. The browser address bar indicates the URL is '44.211.243.234:8081/manage/pluginManager/available'. The left sidebar shows 'Available plugins' is selected. A search bar contains 'Role-based Authorization Strategy'. The main table lists installed plugins with columns: Install, Name, Released, and Health. The 'Role-based Authorization Strategy' plugin (version 918.v91e5468d8db_2) is shown as installed with a green health indicator (100%). Below the table, a description states: 'Enables user authorization using a Role-Based strategy. Roles can be defined globally or for particular jobs or nodes selected by regular expressions.'

The screenshot shows the 'Download progress' section of the Jenkins 'Manage Jenkins' > 'Plugins' page. The left sidebar shows 'Download progress' is selected. The progress section shows the following steps: 'Preparation' (with sub-steps: Checking internet connectivity, Checking update center connectivity, Success), 'Role-based Authorization Strategy' (Success), and 'Loading plugin extensions' (Success). Below the progress list, there are two links: 'Go back to the top page (you can start using the installed plugins right away)' and a checkbox for 'Restart Jenkins when installation is complete and no jobs are running'.

The screenshot shows the Jenkins 'Manage Jenkins' > 'Security' page. The left sidebar shows 'Security' is selected. The page title is 'Security' with a subtitle 'Secure Jenkins; define who is allowed to access/use the system.' Under the 'Authentication' section, there is a checkbox for 'Disable "Keep me signed in"' and a dropdown for 'Security Realm' set to 'Jenkins' own user database'. Under the 'Authorization' section, there is a dropdown for 'Authorization' set to 'Role-Based Strategy'.

← → ↻ Not secure 44.211.243.234:8081/securityRealm/

Jenkins / Jenkins' own user database

Users 3 [+ Create User](#)

	Username 1	Full name	
	Admin	Poojith R	
	DevOps	Devops	
	Testing	Testing	

← → ↻ Not secure 44.211.243.234:8081/manage/configureSecurity/

Jenkins / Manage Jenkins / Security

filter users and groups [add user](#) [add group](#)

DevOps Overall: Read - Job: Build, Cancel, Configure, Create, Delete, Read, Workspace

Overall

Administer Manage [Read](#)

Credentials

Create Delete ManageDomains Update View

Agent

Build Configure Connect Create Delete Disconnect Provision

Job

[Build](#) [Cancel](#) [Configure](#) [Create](#) [Delete](#) [Discover \(implied\)](#) [Move](#)

[Read](#) [Workspace](#)

Run

Delete [Replay \(implied\)](#) Update

View

Configure Create Delete Read

[Save](#) [Apply](#)

← → ↻ Not secure 44.211.243.234:8081/manage/configureSecurity/

Jenkins / Manage Jenkins / Security

filter users and groups [add user](#) [add group](#)

Testing Overall: Read - Credentials: View - Job: Build, Configure, Create, Delete, Read, Workspace

Overall

Administer Manage [Read](#)

Credentials

Create Delete ManageDomains Update [View](#)

Agent

Build Configure Connect Create Delete Disconnect Provision

Job

[Build](#) [Cancel](#) [Configure](#) [Create](#) [Delete](#) [Discover \(implied\)](#) [Move](#)

[Read](#) [Workspace](#)

Run

Delete [Replay \(implied\)](#) Update

View

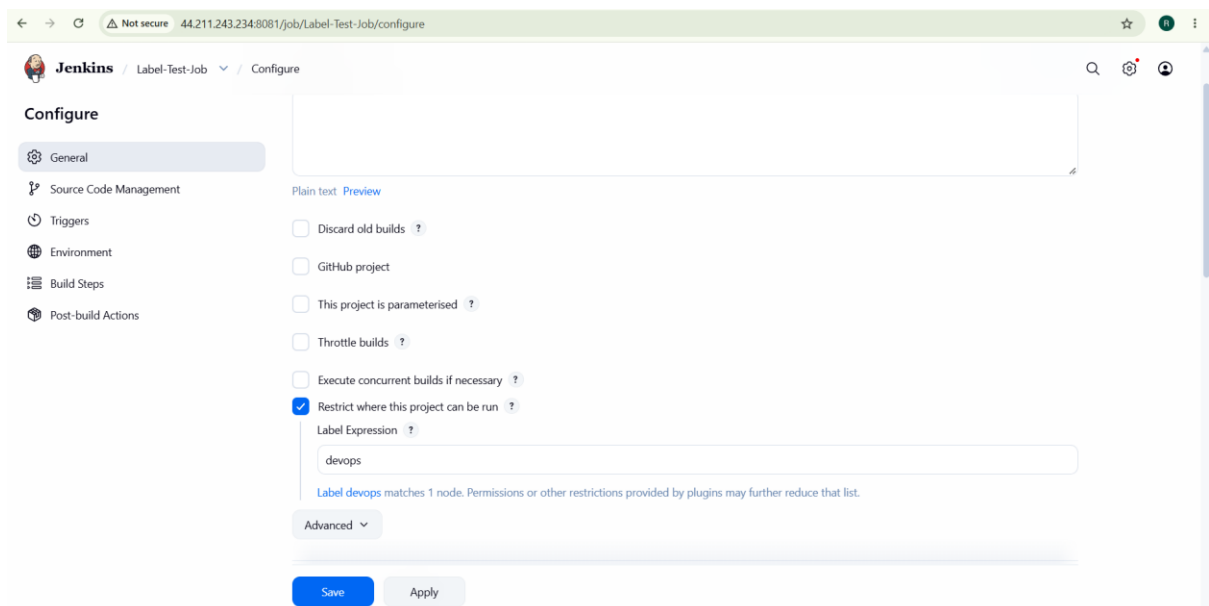
Configure Create Delete Read

[Save](#) [Apply](#)

4. Configure labels and restrict the jobs to execute based on label only.

Implementation Steps

- Open: Jenkins → Manage Jenkins → Nodes
- Open the Built-In Node configuration.
- Add the following labels:
devops
testing
- Save the node configuration.
- The same Built-In Node was used for the training environment, with both labels assigned to it.
- Create a Job Using the devops Label
- Create a Freestyle project:
Label-Test-Job
- Open the job configuration.
- Enable: Restrict where this project can be run
- Enter: devops
- Add a simple build step if required.
- Save the job.
- Click Build Now.
- Verify that the job executes successfully.
- Same for testing label as well



 Jenkins

Label-Test-Job

#1

Search

Settings

Profile

Status

Changes

Console Output

Delete build '#1'

Edit build information

Download

Copy

View as plain text

Started by user Poojith R

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/Label-Test-Job

[Label-Test-Job] \$ /bin/sh -xe /tmp/jenkins13326633019109132698.sh

+ echo 'Running on devops labeled node'

Running on devops labeled node

+ echo Hostname:

Hostname:

+ hostname

ip-172-31-6-74.ec2.internal

+ echo User:

User:

+ whoami

jenkins

Finished: SUCCESS

 Jenkins

Nodes

Built-in Node

Configure

Search

Settings

Profile

Status

Configure

Build History

Load Statistics

Script Console

Build Executor Status

(0 of 2 executors busy)

Number of executors ?

2

Labels ?

devops testing

Usage ?

Use this node as much as possible

Node Properties

☐ Disable deferred wipeout on this node ?

☐ Disk Space Monitoring Thresholds

☐ Environment variables

Save

Apply

 Jenkins

Testing-Label-Job

Configure

Search

Settings

Profile

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Plain text

Preview

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterised ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

testing

Label testing matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced

Save

Apply



5. Create Three sample jobs using the below URL.

Prerequisite – Install Git in your server.

Job 1 — Login-App-Job-1

Implementation Steps

- Create a new Jenkins item: Login-App-Job-1
- Select: Freestyle project
- Enable: Restrict where this project can be run
- Enter the label: devops
- Under Source Code Management, select: Git
- Enter: https://github.com/betawins/Techie_horizon_Login_app.git
- Configure the branch: */master
- No credentials were required because the repository is public.
- Save the configuration.
- Click Build Now.
- Verify that Jenkins successfully clones the repository.

Job 2 — Login-App-Job-2

- Create: Login-App-Job-2
- Select: Freestyle project
- Restrict the job to: testing
- Configure Git:
- Repository: https://github.com/betawins/Techie_horizon_Login_app.git
- Branch: */master
- Add the following build step:
echo "Starting Login-App-Job-2"
echo "Source code checkout successful"
echo "Running on:"
hostname

```
echo "Workspace:"
```

```
pwd
```

- Save the configuration.
- Click Build Now.
- Verify → Finished: SUCCESS

Job 3 — Login-App-Job-3

- Create: Login-App-Job-3
- Select: Freestyle project
- Restrict the job to: devops
- Configure the Git repository:
`https://github.com/betawins/Techie_horizon_Login_app.git`
- Configure the branch: */master
- Add the following build step:

```
echo "Starting Login-App-Job-3"  
echo "Techie Horizon Login App"  
echo "Running on node:"  
hostname  
echo "Build completed successfully"
```
- Save the configuration.
- Click Build Now.
- Verify the console output.

The screenshot shows the Jenkins 'Configure' page for a job named 'Login-App-Job-1'. The left sidebar contains a 'Configure' section with sub-items: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is for 'Source Code Management' and includes the following fields:

- Repository URL**: A text box containing `https://github.com/betawins/Techie_horizon_Login_app.git`. Below it is a red error message: "Failed to connect to repository : Error performing git command: git ls-remote -h https://github.com/betawins/Techie_horizon_Login_app.git HEAD".
- Credentials**: A dropdown menu showing '- none -' and an '+ Add' button.
- Advanced**: A button with a dropdown arrow.
- + Add Repository**: A button.
- Branches to build**: A section with a 'Branch Specifier (blank for 'any')' text box containing '*/master' and a '+ Add Branch' button.

At the bottom of the configuration area are two buttons: 'Save' (in blue) and 'Apply' (in grey).

```
[ec2-user@ip-172-31-6-74 ~]$ sudo su -
Last login: Tue Sep 29 10:19:09 UTC 2026 on pts/1
[root@ip-172-31-6-74 ~]# yum install git
Last metadata expiration check: 1:42:10 ago on Tue Sep 29 10:20:14 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository          Size
-----
Installing:
git                                   x86_64            2.50.1-1.amzn2023.0.1  amazonlinux         53 k
Installing dependencies:
git-core                             x86_64            2.50.1-1.amzn2023.0.1  amazonlinux         4.9 M
git-core-doc                         noarch            2.50.1-1.amzn2023.0.1  amazonlinux         2.8 M
perl-Error                           noarch            1:0.17030-2.amzn2023.0.1  amazonlinux         42 k
perl-File-Find                       noarch            1:27-477.amzn2023.0.9  amazonlinux         26 k
perl-Git                             noarch            2.50.1-1.amzn2023.0.1  amazonlinux         41 k
perl-TermReadKey                     x86_64            2.38-9.amzn2023.0.3    amazonlinux         36 k
perl-lib                             x86_64            0.65-477.amzn2023.0.9  amazonlinux         15 k

Transaction Summary
-----
Install 8 Packages
Total download size: 7.9 M
Installed size: 41 M
Is this ok [y/N]: y
Downloading Packages:
(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm                                1.8 MB/s | 53 kB  00:00
(2/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm                      56 MB/s | 2.8 MB  00:00
(3/8): perl-Error-0.17030-2.amzn2023.0.1.noarch.rpm                       1.4 MB/s | 42 kB  00:00
(4/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm                          58 MB/s | 4.9 MB  00:00
(5/8): perl-File-Find-1.37-477.amzn2023.0.9.noarch.rpm                   740 kB/s | 26 kB  00:00
(6/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm                         1.3 MB/s | 41 kB  00:00
(7/8): perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64.rpm                   1.2 MB/s | 36 kB  00:00
(8/8): perl-lib-0.65-477.amzn2023.0.9.x86_64.rpm                         534 kB/s | 15 kB  00:00
Total                                                                      39 MB/s | 7.9 MB  00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : git-core-2.50.1-1.amzn2023.0.1.x86_64 1/1
                git-core-doc-2.50.1-1.amzn2023.0.1.noarch 1/8
                perl-lib-0.65-477.amzn2023.0.9.x86_64 2/8
                perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 3/8
                perl-File-Find-1.37-477.amzn2023.0.9.noarch 4/8
                perl-Git-2.50.1-1.amzn2023.0.1.noarch 5/8
                perl-Error-1:0.17030-2.amzn2023.0.1.noarch 6/8
                perl-Git-2.50.1-1.amzn2023.0.1.noarch 7/8
                git-2.50.1-1.amzn2023.0.1.x86_64 8/8
Running scriptlet: git-2.50.1-1.amzn2023.0.1.x86_64 8/8
Verifying : git-2.50.1-1.amzn2023.0.1.x86_64 1/8
Verifying : git-core-2.50.1-1.amzn2023.0.1.x86_64 2/8
Verifying : git-core-doc-2.50.1-1.amzn2023.0.1.noarch 3/8
Verifying : perl-Error-1:0.17030-2.amzn2023.0.1.noarch 4/8
Verifying : perl-File-Find-1.37-477.amzn2023.0.9.noarch 5/8
Verifying : perl-Git-2.50.1-1.amzn2023.0.1.noarch 6/8
Verifying : perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 7/8
Verifying : perl-lib-0.65-477.amzn2023.0.9.x86_64 8/8
```

 Jenkins / Login-App-Job-1 / #1

 Status

 Changes

 Console Output

 Delete build #1

 Git Build Data

 Edit build information

Console

 Download  Copy  View as plain text

Started by user [Poojith R](#)
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Login-App-Job-1
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/Techie_horizon_login_app.git
> git init /var/lib/jenkins/workspace/Login-App-Job-1 # timeout=10
Fetching upstream changes from https://github.com/betawins/Techie_horizon_login_app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabir code for testing"
First time build. Skipping changelog.
Finished: SUCCESS

 Jenkins / Login-App-Job-2 / #1

 Status

 Changes

 Console Output

 Delete build #1

 Git Build Data

 Edit build information

Console

 Download  Copy  View as plain text

Started by user [Poojith R](#)
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Login-App-Job-2
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/Techie_horizon_login_app.git
> git init /var/lib/jenkins/workspace/Login-App-Job-2 # timeout=10
Fetching upstream changes from https://github.com/betawins/Techie_horizon_login_app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabir code for testing"
First time build. Skipping changelog.
[Login-App-Job-2] \$ /bin/sh -xe /tmp/jenkins15479645899517295255.sh
+ echo 'Starting Login-App-Job-2'
Starting Login-App-Job-2

```
← → ↻ Not secure 44.211.243.234:8081/job/Login-App-Job-2/1/console
Jenkins / Login-App-Job-2 / #1

> git config remote.origin.url https://github.com/betawins/Techie_horizon_login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabr code for testing"
First time build. Skipping changelog.
[Login-App-Job-2] $ /bin/sh -xe /tmp/jenkins15479645899517295255.sh
+ echo 'Starting Login-App-Job-2'
Starting Login-App-Job-2
+ echo 'Source code checkout successful'
Source code checkout successful
+ echo 'Running on:'
Running on:
+ hostname
ip-172-31-6-74.ec2.internal
+ echo Workspace:
Workspace:
+ pwd
/var/lib/jenkins/workspace/Login-App-Job-2
Finished: SUCCESS
```

← → ↻ Not secure 44.211.243.234:8081/job/Login-App-Job-3/1/console

Jenkins / Login-App-Job-3 / #1

Status

Changes

Console Output

Delete build #1

Git Build Data

Edit build information

Console

Download

Copy

View as plain text

Started by user Poojith R
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Login-App-Job-3
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/Techie_horizon_login_app.git
> git init /var/lib/jenkins/workspace/Login-App-Job-3 # timeout=10
Fetching upstream changes from https://github.com/betawins/Techie_horizon_login_app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabr code for testing"
First time build. Skipping changelog.
[Login-App-Job-3] \$ /bin/sh -xe /tmp/jenkins16591215009662711977.sh
+ echo 'Starting Login-App-Job-3'
Starting Login-App-Job-3

```
← → ↻ Not secure 44.211.243.234:8081/job/Login-App-Job-3/1/console
Jenkins / Login-App-Job-3 / #1

> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabr code for testing"
First time build. Skipping changelog.
[Login-App-Job-3] $ /bin/sh -xe /tmp/jenkins16591215009662711977.sh
+ echo 'Starting Login-App-Job-3'
Starting Login-App-Job-3
+ echo 'Techie Horizon Login App'
Techie Horizon Login App
+ echo 'Running on node:'
Running on node:
+ hostname
ip-172-31-6-74.ec2.internal
+ echo 'Build completed successfully'
Build completed successfully
Finished: SUCCESS
```

6. Create one Jenkins job using GitHub Private repository.

Step 1 — Create GitHub Personal Access Token

- Log in to GitHub.
- Navigate to:
Settings → Developer settings → Personal access tokens → Tokens (classic)
- Generate a Personal Access Token.
- Configure the required repository access.
- For the classic token, enable: repo
- Generate the token.
- Copy and securely store the token.

Step 2 — Create Jenkins Credential

- Open: Jenkins → Manage Jenkins → Credentials
- Select: System → Global credentials
- Click: Add Credentials
- Select:
Kind: Username with password
- Configure:
Username: <USERNAME>
- Password: GitHub Personal Access Token
- ID: github-private-repo
- Description: GitHub Private Repository
- Click Save.

Step 3 — Create the Private Repository Job

- From the Jenkins dashboard, click: New Item
- Enter: Private-Repo-Job
- Select: Freestyle project
- Click OK.

Step 4 — Configure Node Label

- Under General, enable: Restrict where this project can be run
- Enter: devops
- This ensures the job runs on the node configured with the devops label.

Step 5 — Configure Git

- Under Source Code Management, select: Git
- Repository URL: https://github.com/ravutla-10/Private_Repo.git

- Select the Jenkins credential: github-private-repo
- Branch: */main

Step 6 — Add Build Step

- Under: Build Steps → Execute shell
- The following commands were configured:
 echo "Starting Private Repository Job"
 echo "Private repository checkout successful"
 echo "Running on:"
 hostname
 echo "Workspace:"
 pwd
 echo "Repository files:"
 ls -la

Step 7 — Build the Job

- Save the configuration.
- Open: Private-Repo-Job
- Click: Build Now
- Open the build number.
- Select: Console Output
- Verify that Jenkins successfully:
 Authenticated with GitHub.
 Cloned the private repository.
 Checked out the main branch.
 Executed the build commands.
 Displayed the repository files.
 The successful build produced:
 - using credential github-private-repo and
 - Private repository checkout successful
 - The repository file was also displayed:
 - test.txt
 - Finally: Finished: SUCCESS

 Jenkins

Private-Repo-Job

#1

Status

Changes

Console Output

Delete build '#1'

Git Build Data

Edit build information

Console

DownloadCopyView as plain text

Started by user Poojith R
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Private-Repo-Job
The recommended git tool is: NONE
using credential github-private-repo
Cloning the remote Git repository
Cloning repository https://github.com/ravutla-10/Private_Repo.git
> git init /var/lib/jenkins/workspace/Private-Repo-Job # timeout=10
Fetching upstream changes from https://github.com/ravutla-10/Private_Repo.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
using GIT_ASKPASS to set credentials GitHub Private Repository
> git fetch --tags --force --progress -- https://github.com/ravutla-10/Private_Repo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/ravutla-10/Private_Repo.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 6c8144e9dc099afc00ae2050e77f72ce252d7a8a (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 6c8144e9dc099afc00ae2050e77f72ce252d7a8a # timeout=10
Commit message: "Add welcome message to test.txt"
First time build. Skipping changelog.
[Private-Repo-Job] \$ /bin/sh -xe /tmp/jenkins14464074857334632845.sh
+ echo 'Starting Private Repository Job'
Starting Private Repository Job

 Jenkins

Private-Repo-Job

#1

First time build. Skipping changelog.
[Private-Repo-Job] \$ /bin/sh -xe /tmp/jenkins14464074857334632845.sh
+ echo 'Starting Private Repository Job'
Starting Private Repository Job
+ echo 'Private repository checkout successful'
Private repository checkout successful
+ echo 'Running on:'
Running on:
+ hostname
ip-172-31-6-74.ec2.internal
+ echo Workspace:
Workspace:
+ pwd
/var/lib/jenkins/workspace/Private-Repo-Job
+ echo 'Repository files:'
Repository files:
+ ls -la
total 4
drwxr-xr-x. 3 jenkins jenkins 34 Sep 29 12:52 .
drwxr-xr-x. 9 jenkins jenkins 174 Sep 29 12:52 ..
drwxr-xr-x. 7 jenkins jenkins 146 Sep 29 12:52 .git
-rw-r--r--. 1 jenkins jenkins 38 Sep 29 12:52 test.txt
Finished: SUCCESS